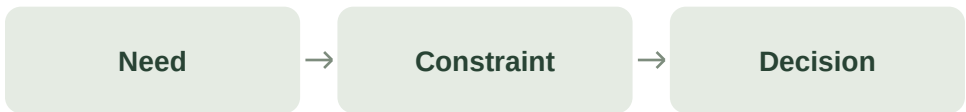


System Design

Start with the job

Before drawing components, establish who uses the system, what they need to do, and what failure would mean. An architecture solves a particular set of constraints.

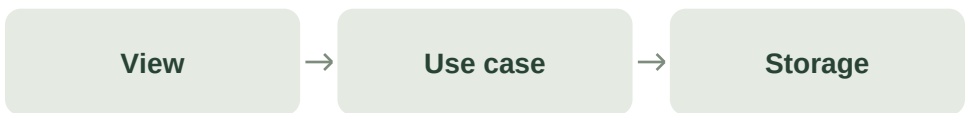


For a personal PDF reader, local access, data integrity and fast navigation may matter more than multi-region availability. Do not pay for complexity that the task does not require.

A useful question: what must still work when the internet does not?

Make data ownership clear

Give each responsibility an owner. The interface presents state. An application service coordinates a workflow. Persistence commits durable changes.



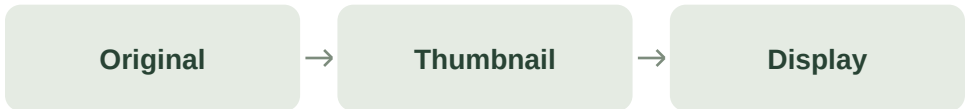
Passing a database object through every screen makes boundaries difficult to change. Passing an immutable document summary keeps rendering separate from storage details.

Identify which layer should know about file paths, and which should only know about a document identifier.

Caching is a tradeoff

A cache avoids repeating work, but introduces a second representation that can become stale. Decide what is authoritative, how entries are invalidated and how much memory the cache may use.

PDF thumbnails are derived data. If a thumbnail disappears, recreate it. Original documents and private annotations are not disposable caches.



Ask: can this information be recreated without asking the user to do the work again?

Design the failure path

An import is not complete when a preview appears. It is complete when the required data has been safely stored and the operation has been acknowledged.

A recoverable design considers full storage, interrupted writes, duplicate requests and corrupted input. Error messages must tell the user what happened without claiming that missing work was saved.

This is original sample material illustrating general engineering reasoning. The architecture and recovery semantics of this app are documented in the source package.

Review exercise: sketch a save operation that can fail at any step without overwriting the original.